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BHATTADEV UNIVERSITY

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BHATTADEV UNIVERSITY IN PARTIAL FULLFILMENT OF THE
REQUIREMENT FOR THE AWARD OF DEGREE-BACHELOR OF ARTS
(ECONOMICS)

TITLE OF THE PROJECT REPORT

**Comparative Study of Traditional vs. Modern Dairy
Farming Practices**

PAPER: ECO-HE-6026-THE ECONOMY OF ASSAM

SUBMITTED BY

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6TH SEM, B.A (ECONOMICS)
REGISTRATION NO:
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UNDER THE SUPERVISION

SOMEONE
DEPARTMENT OF ECONOMICS

UNDERTAKING THE PROJECT REPORT
ON
BACHELOR OF ARTS (ECONOMICS)

TOPIC OF THE PROJECT REPORT

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I hereby declare that above this work has been done by me and has not been submitted elsewhere.

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CERTIFICATE

I Certify that project report work entitled, “**Comparative Study of Traditional vs. Modern Dairy Farming Practices**”. A CASE STUDY ON SOME SELECTED SHGS IN BAJALI submitted by Meri pasandida Aurat.

It is her own work and has been done in the light of evaluator’s comments under my supervision.

It is recommended that this project be placed before the examiner for evaluation.

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INTRODUCTION

1.1 Overview of Dairy Production

Dairy farming plays a vital role in the rural economy of Assam, particularly in the Bajali Area, where it serves as a primary livelihood for many households. Traditionally, dairy farming in this region has been characterized by conventional methods involving indigenous cattle breeds, manual milking, and basic feeding practices. However, with the advent of modern agricultural techniques, there has been a gradual shift towards more efficient, technology-driven dairy farming practices.

Modern dairy farming practices in Bajali areas include the use of crossbred cattle, automated milking systems, improved feeding techniques, and scientific herd management. These practices have significantly impacted milk productivity, animal health, and farmers' economic conditions. In contrast, traditional methods, despite being less efficient, hold cultural significance and require lower initial investments.

This comparative study aims to analyse the differences between traditional and modern dairy farming practices in the Bajali area of Assam. It will evaluate the economic, social, and environmental impacts of both approaches while highlighting the challenges and opportunities associated with their adoption. Through this study, we seek to provide insights into how modernization affects dairy farming and the potential for sustainable development in the region.

1.2 Aim of Study

The primary aim of this study is to conduct a comprehensive comparison between traditional and modern dairy farming practices in the Bajali region of Assam. The study intends to examine how each approach influences milk production, income generation, animal welfare, and sustainability. By exploring the economic, technological, and social dimensions of dairy farming, the research seeks to identify the strengths and limitations of both traditional and modern methods. Additionally, this study aims to assess how modernization is reshaping the rural dairy sector and to offer practical recommendations for improving dairy productivity and livelihoods while preserving valuable traditional knowledge.

1.3 Area of Study

The study is conducted in the **Bajali district** of Assam, a region known for its agrarian economy and significant dependence on dairy farming as a source of livelihood. Bajali, a newly formed district carved out of Barpeta in 2021, comprises both rural and semi-urban areas where traditional and modern dairy practices coexist. The region presents a suitable setting for comparative research due to its mix of indigenous farming communities and progressive dairy farmers who have adopted modern techniques. Specific villages and dairy clusters within Pathsala and surrounding areas such as Milanpur, Titka, Rehabari, and Nityananda have been selected for detailed observation and data collection. This geographical scope offers a balanced view of the transition from traditional to modern dairy systems in a real-world rural context.

Review of Literature

A literature review provides an overview of the existing research on a particular topic. It helps to identify what is already known, what gaps exist, and how the current study fits into the broader academic conversation. In this review, I will summarize and analyze key studies related to "The Comparative Study of Traditional Vs Modern dairy farming practices" highlighting major findings, ongoing debates, and areas where further research is needed.

1. **Title:** Impact of dairy farming for changing livelihood through AACP intervention in Bajali area of Barpeta District, Assam.

Author: Ira Das et al.

Published: 2014

Dairy farming is a part of the livestock sector and can play a significant role in sustaining rural livelihood in developing countries like India. Assam, in the North Eastern Region of India, is a backward state compared to other states of India. Assam has the lowest per capita availability of livestock products. However, a large quantity of milk, eggs and meat products are imported from other parts of the country to meet the state's requirement. Recently, there is a progressive development of dairy activities in and around Bajali Development Block of Barpeta District leading to substantial increase in milk production through formation of Dairy Co-operative Societies during Assam Agricultural Competitiveness Project (AACP). Therefore, in this study, an attempt has been made to examine the impact of dairy farming for changing livelihood during AACP intervention in Bajali area of Barpeta District. It has been found from the study that there is significant impact of Dairy Farming for Changing Livelihood during AACP intervention in Bajali area of Barpeta District. The socio-economic impact of the project on the livelihood of the dairy farmers is prominent in the study area. In the long run, dairy farming can help poverty reduction and can ensure sustainable livelihood for the people of the study area if proper attention is paid to the problems faced by the farmers.

2. Title: Supply Chain Analysis in Milk Marketing: A Comparative Study of DUSS

Member and Non-Member Dairy Farmers in Barpeta District of Assam

Author: Jonali Sarma and R N Barman

Published: 2014

In Assam milk marketing has been largely unorganised. Raw milk is the primary dairy product marketed in the state. Over 90 per cent of the unprocessed milk is collected and marketed through the informal markets. In some selected pockets of the state including Barpeta district where the present study has been undertaken there have been involvement of co-operative societies in the milk marketing channels. The study attempted to assess the consumption and disposal pattern of milk along with the estimation of marketing costs, marketing margins, price spread and marketing efficiency of milk in different milk supply chains in Barpeta district of Assam which had the highest number of livestock population among the districts of Assam as per 2003 Livestock Census. For the purpose data pertaining to the year 2012-13 were collected through schedules and questionnaires from the sampled farmers and cooperative societies of Bajali and Bhawanipur development blocks of the district to assess the performance of the member and non-member farmers of Dugdha Unnayan Samabai Sammittees (DUSS). The study indicated that the proportions of local breeds were more in the small size of farms as compared to medium and large size groups of farms. The percentage of cross breed jersey cows was higher as compared to Holstein and local breeds in all the villages and across all size groups of farms. Demand for cross breed Jersey cows have been increasing because of higher milk yield and acceptable fat and SNF contents in the milk. Among the non-members the small farmers largely sold milk directly to their consumers and only 21.81 percent sold milk to restaurants and other retail outlets. Among the medium and large farmers majority of the farmers (38.46 per cent) distributed milk through milk vendors, 33.84 per cent sold milk to milk retailers and 27.69 per cent sold milk to different hotels and restaurants. The member farmers of DUSS directly sold their marketable surplus to their respective societies and the societies in turn sold a major portion to the West Assam Milk Union Ltd. (WAMUL) run major milk plant 'Purabi dairy' at Guwahati and a minor portion to "Asomi dairy plant" at Sarbhog in the district from where after proper processing and packaging the milk reached the consumers' hands through various retail outlets. The village level co-operative societies sold some portion to local consumers through their sales counters. Of the total five marketing channels for the non-member farmers categories and two marketing channels for the DUSS member farmer categories were identified and

analysed. On the whole the DUSS in the Bajali development block provided on an average 1849.08 days of total employment of which 1505.45 man-days of full time annual employment and 343.43 man-days of part time annual employment. In the Bhawanipur development block the employment generation by the milk cooperative societies was on an average of 1230 man-days per year. The DUSS involvement in the milk marketing in the district streamlined the milk supply network in the area.

3. Title: Dairy Cattle Management and Marketing of Milk in Lower Brahmaputra Valley Zone of Assam.

Author: Mukut Das

Published: 2014

Two hundred dairy farmers having at least two milch cows were selected randomly for collection of data. A structured, reliable and valid interview schedule was prepared, pre-tested and was employed for data collection. All the obtained data were tabulated and put to standard statistical test to facilitate proper interpretation and to arrive at some conclusions as per the pre-formulated objectives of the study. The study revealed that the overall average age at first calving, lactation yield, lactation length, intercooling period, dry period and service period in crossbred cattle were found to be 31.03 ± 0.09 months, 1441.18 ± 26.53 litres, 292.32 ± 0.72 days, 385.64 ± 0.85 days, 93.32 ± 0.99 days and 111.95 ± 0.85 days respectively in crossbred cattle irrespective of locations, while the corresponding values for indigenous cattle were recorded as 42.34 ± 0.10 months, 280.75 ± 7.45 litres, 238.60 ± 1.83 days, 538.42 ± 1.69 days, 299.86 ± 2.03 days, 258.57 ± 1.71 days respectively. All the economic traits were found to be significantly influenced by the genetic group and the location except for the lactation yield within the location in indigenous cattle. In regards to the housing of cattle in the study areas, the dairy farmers used 56.84 and 43.16 per cent thatch and corrugated iron sheet roof respectively in crossbred cattle. However, majority of the dairy farmers (81.48%) and only 18.52 per cent used corrugated iron sheet as roof material for housing of the indigenous cattle. Most of the dairy farmers (47.95%) had cattle shed with half wall, 23.97 per cent with full wall and 28.08 per cent without wall in respect of crossbred cattle. The corresponding data in respect of indigenous cattle were 25.93, 62.96 and 11.11 per cent respectively in the study areas.

4. **Title:** Assessment of the Status of Dairying and Potential to Improve Socio-Economic Status of the Milk Producers and Convergence of all Central and State Schemes at District Level in Assam.

Author: Gautam Kakaty, Anup K Das

Published: 2017

The dairy farming is an important enterprise that provides employment, income and nutritive food for millions of rural families and also supplies cow dung as organic manure to enrich the soil fertility and thus helps in increasing crop production. Assam, with vast natural endowment, has the enormous potentiality for the development of dairy sector. The study covers 120 milk producers from DCS and another 120 from NDCS of four different districts, viz. Barpeta, Jorhat Kamrup and Nagaon in Assam. The findings of the study show that although the status of dairying in Assam is far from satisfactory level, there lies ample opportunities which still remain un realized because of a plethora of problems and difficulties. Optimum utilization of natural resources, creation of adequate health-care facilities for livestock, improvement of breeding programmes through Artificial Insemination, improvement of present milk marketing system and timely vaccination can go a long way in bringing marked changes in the lives of the milk producers of this part of the country.

5. **Title:** Attributes Influencing Adoption of Health Care Scientific Dairy Management Practices among Tribal and Non-Tribal Dairy Farmers of Assam.

Author: Ram Uma Tamuli et al.

Published: 2021

The study was carried out to ascertain attributes (variables) influencing adoption of health care scientific dairy management practices among the tribal and non-tribal dairy farmers of Assam. Descriptive (survey method) was used as the study design. The study was conducted in the tribal and non-tribal villages of Sonitpur, Lakhimpur and Dhemaji district of Assam for a duration of 6 months during October, 2019 to March, 2020. A total of 240 (120 each tribal and non-tribal dairy farmers) numbers of respondents were selected for the study. Data were personally collected by interviewing the respondents with an interview schedule. It was observed that experience in dairy farming, extension contact, dairy farming attitude and aspiration level were identified as

crucial in their contribution to total adoption and explained about 71 percent variations in the total adoption in case of tribal dairy farmers.

6. **Title:** Study on socio economic status of dairy farmers and its impact on environment at Topatoli village of Kamrup Metropolitan district of Assam, India.

Author: Bhaskar Dev Swargeary, Durlov Jyoti Kalita

Published: 2023

This study was conducted to analyze the socio-economic status of dairy farmers and impact of dairying practice on the environment at Topatoli village of Kamrup (M) district of Assam, India. The research employed both quantitative and qualitative methods using the questionnaire method to answer the research questions and structured face to face in-depth interviews to complement the quantitative findings. In the study area out of 21 respondents, 90.48% dairy farmers were primarily involved in dairying practice and the only source of their livelihood. The respondents with income > 50000 thousand per month constitute 14.28%. Income between 21000-30000 and 31000-40000 constitute the same percentage that is 9.52%. The highest percentage of respondents with 52.38% with range between 11000-20000 per month.

2.1 Research Gap

While several studies have explored various aspects of dairy farming in Assam—such as socio-economic impacts, marketing efficiency, housing practices, breed performance, and health management—there is a notable lack of comparative analysis between traditional and modern dairy farming practices in terms of sustainability, productivity, and adoption barriers. No study comprehensively evaluates how these two systems differ in practical outcomes, especially across diverse socio-economic and geographic contexts in Assam. This leaves a gap for research that directly compares the two to inform policy and practice.

2.2 Objectives of the study

- To compare the socio-economic impacts of traditional and modern dairy farming practices on rural livelihoods in selected districts of Assam.
- To assess the adoption level and influencing factors of scientific dairy management practices among tribal and non-tribal farmers in Assam.

Methodology of the Study

3.1 Significance of the study area

The **Bajali district** holds significant importance for this study due to its strong dependence on agriculture and dairy farming as a means of livelihood. Many households in this region are engaged in small-scale dairy activities, making it an ideal location to observe both traditional and modern dairy practices side by side. The area reflects a transitional phase where some farmers continue with age-old methods while others are gradually adopting modern techniques like improved cattle breeds.



Studying Bajali provides valuable insights into how modernization is influencing rural dairy farming in Assam. It also helps in understanding the challenges that traditional farmers face in adapting to new technologies and the benefits experienced by those who have already made the shift.

3.2 Data Collection

This study is based on both **primary** and **secondary data** sources to ensure a comprehensive and reliable analysis of traditional and modern dairy farming practices in the Bajali district of Assam.

Primary Data:

Primary data was collected directly from dairy farmers in various panchayat of the Bajali district through a structured questionnaire. The survey included both open-ended and close-ended questions focusing on key aspects such as:

- Type of farming practice (traditional/modern)
- Cattle breeds and milk yield
- Feeding methods and animal healthcare
- Milking and storage systems
- Marketing and income patterns
- Challenges faced by farmers

Respondents were selected using a **random sampling method**, ensuring representation from both traditional and modern farming households. The data was gathered through **personal interviews and field visits**.

Secondary Data:

Secondary data was gathered from credible sources such as:

- Research journals
- Previous studies on dairy farming in Assam and India
- Various articles from credible sources

These sources helped in understanding broader trends, cross-verifying primary data, and providing context for the comparative analysis.

3.2 Sample design

The samples for the study are collected from some areas of Bajali District. The collected sample size is 30. To analyze the goal of the scheme we have taken two villages, such as

QUESTIONNAIRE

SHG Member's Personal Information:

1. Personal Information:

- Name of the respondent:
- Age:
- Gender:
- Address:

2. How long have you been involved in dairy farming?

3. Which type of dairy farming do you follow? –

Traditional ☐ Modern ☐ Both ☐

4. What are the primary breeds of cattle you keep?

Indigenous ☐ Crossbreed ☐ Exotic ☐

5. What is the average milk yield per cow per day (in litre)?

6. What type of fodder do you like?

Natural grass ☐ Cultivated fodder ☐ Commercial feed ☐

Mixed. ☐

7. Do you follow any scientific feeding practices?

Yes ☐ No ☐

8. Do you provide mineral supplements to your cattle?

Yes ☐ No ☐

9. How do you manage animal health and hygiene?

Traditional methods ☐ Veterinary care ☐ Both ☐

10. What is the method of milking?

Manual ☐ Machine ☐

11. How do you store the milk after collection?

Traditional Vessels ☐ Refrigeration ☐

Cooperative / Collection centre ☐

12. What is your primary mode of selling milk

Direct to customers ☐ Middlemen ☐

Dairy cooperative ☐ Local markets ☐

13. What are the main challenges you face in dairy farming?

Low productivity ☐ Disease management ☐ Lack of

modern equipment ☐ market accessibility ☐

financial constraints. ☐

14. Have you received any training or support from government agencies or NGO's?

Yes ☐ No ☐

15. Are you aware of modern dairy farming techniques?

Yes ☐ No ☐

16. Do you think modern dairy practices are more beneficial compare to traditional ones?

Yes ☐ No ☐ Not Sure ☐

17. Are you planning to upward your dairy farming practices?

Yes ☐ No ☐ May be ☐

18. What is your average monthly income from dairy farming?

19. In your opinion, which farming practices is more sustainable in longer run?

20. Do you plan to continue dairy farming in the future? Why or why not?

21. What improvements would you like to see in dairy farming practices?

22. How many milking cows do you currently have?

23. Do you practice food supplementation with minerals and vitamins?

24. What type of cattle housing do you have?

Modern shed ☐ Open barn ☐ etc ☐

25. Modern shed open barn etc?

26. Do you have separate area for milking, feeding and resting?

27. How often do you conduct routine health check-ups?

28. Do you have access to modern veterinary services in Bajali?

29. In your experience, which practice (traditional or modern) results in higher milk yield per cow, and why do you think so?

30. In your experience, which method requires more labour:

Traditional or modern dairy farming?

30. What are the key differences in cattle health management between traditional and modern practices?